

Geometry-Guided Generalizable FER: Anatomical Attention Aligns Mechanistically but Improves Accuracy Conditionally

Jin Ho Kwak Amsyar Hakimi Chew Glenn
Korea University

Abstract

We study whether a fixed facial-anatomy prior improves zero-shot cross-dataset facial expression recognition (FER). We start from CAFE, which already learns sigmoid channel masks over fixed face features using channel separation and diversity losses. The only added component is a deterministic spatial prior. Specifically, detected 68-point landmarks are mapped through a FACS codebook to expression-wise Action Unit (AU) heatmaps. These heatmaps are then applied to the corresponding CAFE channel groups. The intervention works mechanistically: AU specialization rises from nearly zero to 0.183–0.188 and activations become visibly more localized. However, this large internal shift is not matched by a consistent accuracy gain. Across five source datasets, augmentation is the reliable driver (+3.11 pp), while FACS attention alone is marginal (+0.21 pp) and source-dependent. A per-channel variant further degrades accuracy, showing that where the prior is injected matters. These single-seed diagnostic results suggest that anatomical alignment is useful for mechanistic interpretation but does not by itself ensure robust cross-dataset FER.

1 Introduction

Generalizable FER evaluates a model trained on one source dataset on unseen target datasets without target-domain fine-tuning [1]. This is difficult because RAFDB, FERPlus, AffectNet, SFEW, and MMA differ in identity, resolution, illumination, pose, and label noise. CAFE addresses the setting by keeping large-model face features fixed and learning sigmoid masks that select expression-related channels; it also splits channels into seven expression groups and regularizes them with separation and diversity losses [1]. These are original CAFE components. Our question is narrower: can CAFE benefit from an explicit anatomical *spatial* reference?

The motivation is that learned spatial activations can remain broad over the face, while expression evidence is tied to relatively stable facial structures such as brows, eyes, cheeks, and mouth. We therefore use a fixed landmark/FACS codebook. “Fixed” means that the semantic rule expression $\rightarrow$ AU $\rightarrow$ landmark region is fixed, not that pixel positions are

fixed: landmarks are detected separately for each face, so the heatmap follows pose and scale while preserving anatomical meaning. This design is plausible for cross-dataset transfer because dataset-specific appearance changes, whereas facial-part semantics are preserved: mouth regions remain mouth regions and brow regions remain brow regions. Unlike prior AU-cue work that primarily uses spatial AU cues to improve or interpret FER predictions [3], our goal is diagnostic: after inserting the same type of prior into CAFE, does stronger AU alignment actually translate into cross-domain accuracy improvement?

Our findings lead to a deliberately modest conclusion. The fixed FACS prior reliably induces the intended AU-specialized alignment, but this internal change produces only small and source-dependent accuracy gains. In contrast, augmentation improves cross-dataset accuracy across all five source settings. We therefore frame the result as a mechanism–performance asymmetry: anatomical attention changes where the model attends, but it is not a reliable driver of cross-domain accuracy on its own. Our diagnostic contribution is to separate these two effects in CAFE: the anatomical prior changes the representation, while augmentation remains the more dependable transfer lever.

2 Method

CAFE baseline. We use CAFE without changing its main training objective [1]. A frozen CLIP visual encoder produces a 512-d face feature [2]. A ResNet-18 FER branch [6] produces a spatial feature map $F \in \mathbb{R}^{512 \times 7 \times 7}$, which is pooled and passed through a sigmoid to form a channel mask $m \in [0, 1]^{512}$. The selected feature is $z = c \odot m$. The 512 channels are partitioned into seven expression groups and optimized with

$$\mathcal{L} = \mathcal{L}_{cls} + 1.5\mathcal{L}_{sep} + 5.0\mathcal{L}_{div}, \quad (1)$$

where $\mathcal{L}_{sep}$ encourages class-specific groups and $\mathcal{L}_{div}$ encourages diverse active channels.

Per-group FACS attention. For each image, we detect 68 facial landmarks with a pretrained face-alignment network [5]. A FACS codebook [4] maps each expression to active AUs and the corresponding landmark regions. We ras-

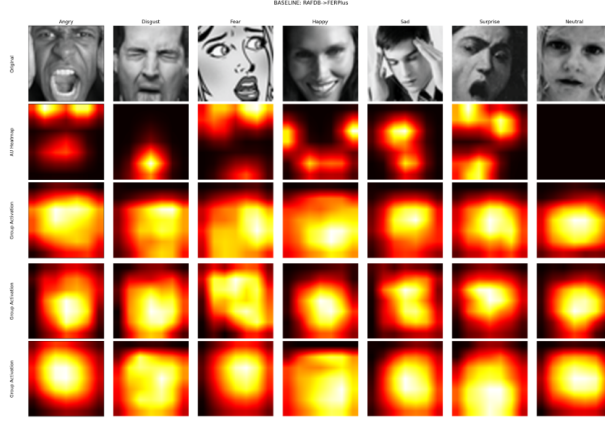


Figure 1. Motivation. Baseline CAFE group activations are broad and diffuse over much of the face (average cosine similarity to AU references: about 0.63), motivating a fixed landmark/FACS spatial reference.

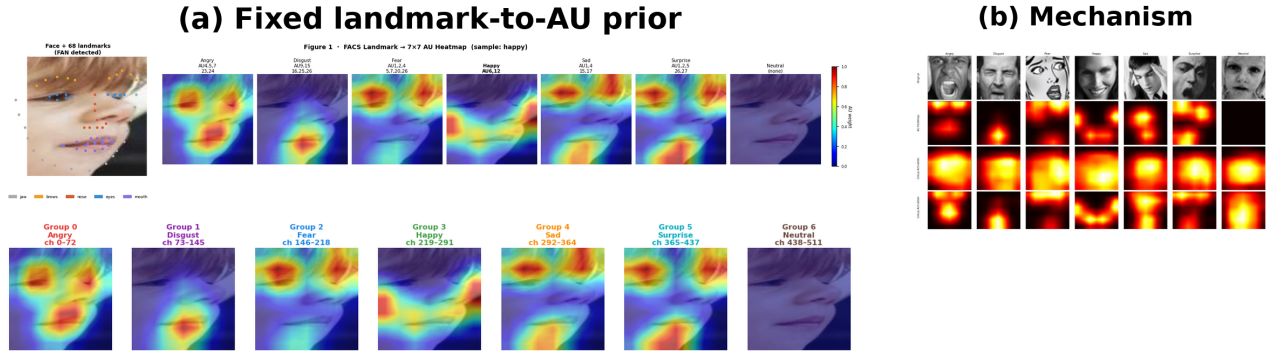


Figure 2. Anatomical prior and mechanism. (a) Detected landmarks are mapped to expression-wise 7×7 AU heatmaps and assigned to CAFE channel groups. (b) Row 2 is the landmark/AU heatmap, row 3 is the original CAFE activation, and row 4 is the AU-gated activation. The gate localizes groups anatomically, but this mechanism does not guarantee a corresponding accuracy gain.

terize the selected points onto a 7×7 grid, apply Gaussian smoothing, and normalize to $[0, 1]$ to obtain an expression heatmap H_k . Landmark detection failed on an average of 8% of samples; those cases, and the neutral class, use a uniform map. For the CAFE channel group G_k assigned to expression k , we apply

$$\tilde{F}_{G_k} = F_{G_k} \odot H_k, \quad k = 1, \dots, 7, \quad (2)$$

then continue with the original CAFE pooling and mask pathway. The gate adds no learnable spatial parameters. This is intentionally less flexible than learned landmarks or a learned spatial mask: a learned prior can adapt to source-specific texture, pose, or annotation shortcuts, whereas the fixed codebook preserves the same anatomical semantics at test time. The trade-off is that the prior is coarse and can suppress useful non-AU cues; the results below test whether the anatomical constraint is worth this capacity cost.

Negative control: per-channel injection. We also tested a channel-wise AU weighting variant, motivated by the idea that more landmark-derived information might help. It reduced in-domain accuracy by about 4.5 pp and cross-domain

accuracy by about 7.6 pp. This supports a design lesson: channel weighting suppresses many channels that CAFE’s separation/diversity losses try to keep active and diverse, so the injection axis can reverse the effect of the same prior.

3 Experiments

We train on one source among RAFDB [8], FERPlus [9], AffectNet [10], MMA (following the CAFE benchmark [1]), and SFEW [11], then evaluate on all five datasets. Cross-domain accuracy averages the four non-source targets. We compare four settings: **Baseline**, **+Aug**, **+FA**, and **+FA+Aug**. Augmentation is implemented with Albumentations [7] and includes horizontal flipping, rotation, cropping, color jittering, and Gaussian noise. Landmarks are re-detected after augmentation so the AU map stays aligned.

To quantify the intended mechanism, we compute a specialization matrix M by averaging over all evaluated samples from each source-trained model. Entry M_{ij} is the cosine similarity between expression group i ’s activation and AU

Table 1. Source-excluded cross-domain changes and AU specialization. Gains are percentage points. $\text{Aug}\Delta = \text{BASELINE_Aug} - \text{BASELINE}$; $\text{FA}\Delta = \text{PerGroup_FA} - \text{BASELINE}$; $\text{FA}\Delta \text{ w/ Aug} = \text{PerGroup_FA_Aug} - \text{BASELINE_Aug}$.

Source	Base	Aug Δ	FA Δ	FA Δ w/ Aug	FA Gap	FA+Aug Gap
RAFDB	52.44	+4.84	+1.02	-0.61	0.228	0.239
FERPlus	51.72	+2.74	-3.26	+3.95	0.215	0.227
AffectNet	35.36	+3.25	+2.00	+2.05	0.195	0.191
MMA	61.19	+1.10	-0.47	-0.52	0.202	0.197
SFEW	32.48	+3.63	+1.77	-0.71	0.077	0.086
Mean	46.64	+3.11	+0.21	+0.83	0.183	0.188

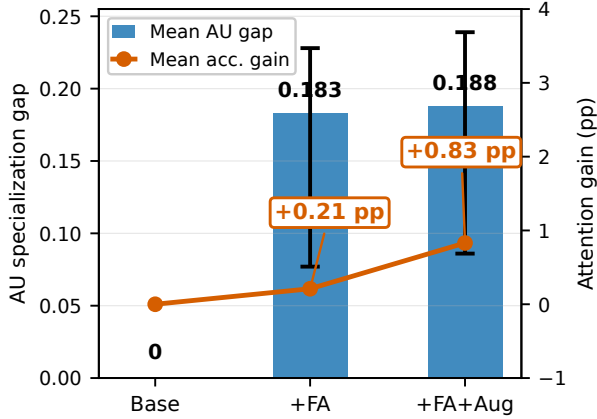


Figure 3. Mechanism–performance asymmetry. FACS attention shifts the intended AU gap from nearly zero to about 0.18–0.19, while the mean accuracy gain remains small. Whiskers on the blue bars show source-wise min/max values for the AU gap; orange labels show the mean accuracy gain in percentage points.

heatmap j . We report

$$\text{gap} = \text{mean}(\text{diag}(M)) - \text{mean}(\text{offdiag}(M)). \quad (3)$$

A larger positive gap means each group aligns more with its own expression AU region than with other regions. Since experiments are single-seed, we avoid significance claims and interpret the numbers as diagnostic evidence.

4 Results and Discussion

Augmentation is the reliable driver. Table 1 shows that augmentation improves all five sources (+1.10 to +4.84 pp; mean +3.11 pp). This is the strongest quantitative pattern. FACS attention is much smaller: +0.21 pp alone and +0.83 pp on top of augmentation.

Mechanism changes more than accuracy. The AU gap rises from approximately zero to 0.183 with FACS attention and 0.188 with FACS attention plus augmentation; Figure 2 also shows more localized activations. Yet this internal shift is not matched by consistent transfer gains. A useful interpretation is a robustness–capacity trade-off: fixed anatomical priors preserve a stable semantic reference for zero-shot transfer, but can suppress non-AU regions that still carry

Table 2. Per-channel AU injection as a negative control. This ablation was run separately from Table 1, with its own baseline and augmentation baseline. Absolute Baseline/+Aug values should therefore be compared within this table, not matched numerically to Table 1. The small baseline drift across runs also supports our conservative treatment of sub-1 pp effects.

Setting	In-domain	Cross-domain	Cross-domain Δ
Baseline (chan. run)	70.1	46.1	–
+Aug (chan. run)	70.6	50.7	+4.6
+ChannelW_FA	65.7	39.1	-7.0 vs base
+ChannelW_FA+Aug	66.0	43.0	-7.7 vs aug

useful expression evidence. Thus, anatomical alignment alone is insufficient for reliable accuracy improvement; we do not treat the AU gap as a statistically validated predictor of accuracy.

Source-wise observations. Source-specific labels are illustrative rather than formal regimes. AffectNet is consistently positive, likely because a noisy in-the-wild source leaves room for residual geometric structure. FERPlus shows a rescue-like pattern: attention alone appears overspecialized to its clean, low-resolution source statistics, but augmentation broadens the source and makes the prior useful. RAFDB and SFEW appear redundant with augmentation, where appearance invariance is already supplied. MMA is accuracy-insensitive: all four cross-domain scores lie in a 60.72–62.29 band despite AU gap = 0.202.

Why the negative control matters. Table 2 shows that channel-wise AU injection is harmful within its own run. Because this ablation used a separate notebook and baseline, we interpret it using within-run deltas. The ~ 0.5 – 1.0 pp baseline drift warns us not to overread sub-1 pp effects in Table 1, but the channel variant’s ~ 7 pp drop is too large to explain by that drift. The same anatomical signal can be compatible with CAFE before pooling, but destructive when it reweights the channel vector that CAFE tries to separate and diversify.

Difference from prior AU-cue FER. Prior AU-cue methods mainly use AU maps for interpretability or prediction within their own protocols. Our question is narrower: in CAFE’s zero-shot cross-dataset setup, does AU alignment improve transferable accuracy? The answer is conditional: the prior is mechanistically valid, but augmentation is the stronger performance lever.

Transfer difficulty and capacity cost. The source-level averages in Table 1 hide a useful target-level pattern. Under no augmentation, we compute $\Delta_{\text{NO}} = \text{Acc}(\text{PerGroup_FA}) - \text{Acc}(\text{BASELINE})$ for all source-target transfers. We then exclude the two lower-tail observations identified by the $1.5 \times \text{IQR}$ rule over Δ_{NO} and group the remaining transfers by baseline transfer accuracy (Table 3). Attention is more useful when the target is hard and becomes negligible on easy transfers.

Table 3. Attention effect by transfer difficulty without augmentation. Tier means use the 18 remaining transfers after excluding two lower-tail outliers by the $1.5 \times \text{IQR}$ rule on Δ_{NO} : FERPlus→SFEW (-8.58 pp) and FERPlus→RAFDB (-6.98 pp).

Tier	n	Baseline transfer acc.	Mean FA effect
Hard	6	18.8–38.7	+1.65 pp
Medium	6	38.8–49.8	+1.42 pp
Easy	6	51.7–75.0	-0.10 pp

This supports the robustness–capacity interpretation: after the criterion-based exclusion in Table 3, anatomical gating is most helpful on hard transfers, but it can also discard useful non-AU cues when the baseline already transfers well.

Design implications. The results suggest three reporting rules for geometry-guided FER. First, report mechanism metrics and task metrics separately; a higher AU gap is not itself a performance claim. Second, compare anatomical priors against a strong augmentation baseline, since augmentation supplies the most consistent domain variation. Third, match the injection site to the backbone: spatial gating before pooling is compatible with CAFE, while channel-wise weighting conflicts with the separation/diversity objective.

5 Conclusion

This single-seed diagnostic study shows that fixed FACS attention is a valid mechanism for inducing AU-aligned specialization in CAFE, but that alignment is not sufficient for reliable cross-dataset accuracy improvement. The main evidence is asymmetric: augmentation improves every source setting, the AU gap shifts by about 0.18, and channel-wise injection drops cross-domain accuracy by about 7 pp. Source-wise labels such as rescue, redundancy, and insensitivity should therefore be read as explanatory observations rather than stable regimes.

The practical conclusion is that anatomical priors should be evaluated as incremental cues on top of a strong augmentation baseline, not as replacements for domain-diverse training. Their value depends on transfer difficulty: the stable semantic reference can help on harder transfers, while suppressing non-AU regions introduces a capacity cost. Future work should verify the observed patterns with repeated seeds, per-dataset landmark reliability statistics, and finer AU codebooks.

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